Mathematics for Derivatives Pricing

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$$\mathbb{E}[\varphi(X)] \geq \varphi(\mathbb{E}[X])$$

$$\int_a^b f(x)dx = F(b) - F(a)$$

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Introduction

Learning derivatives pricing is not an easy task. Aside all the finance knowledge required to understand the products, we also need to have a good grasp of many branches of mathematics, such as:

- **Probability and Stochastic Calculus**: To model the randomness in asset prices and to derive pricing formulas.
- **Calculus**: To compute sensitivities, solve differential equations, and perform optimization.
- **Numerical Methods**: To implement and solve models that may not have closed-form solutions.
- **Linear Algebra**: For high-dimensional problems, that involve multiple assets or factors.

Introduction

For anyone trying to break into the field of derivatives pricing, it might be overwhelming to know where to start.

For this reason, the aim of this presentation is:

1. To provide a clear and concise introduction to the mathematical concepts that are essential for derivatives pricing.
2. To focus on the intuition and practical applications of these concepts, rather than on rigorous mathematical proofs.
3. To equip you with the tools and understanding needed to start learning about derivatives pricing and to build your own models.

It is a lot of material to cover, but we will go through it step by step, so

Don't Panic!

Probability

Probability is the foundation of derivatives pricing, as it allows us to model the uncertainty and randomness in financial markets.

In derivatives pricing, we often use probability to:

- Model the behavior of asset prices over time.
- Calculate expected values of future payoffs.
- Derive pricing formulas.

Pricing a financial instrument is essentially **calculating the expected value of its future payoff**, discounted to the present.

Probability

How can we model the future value of asset prices, interest rates, or other financial variables? A common approach is to use **random variables** to represent these future values.

Random Variables

A continuous random variable X is a variable that can take on different values based on the outcome of a random event. It can be characterized by two main functions:

- **Probability Density Function (PDF)**: denoted by $f_X(x)$, it describes the likelihood of the random variable taking on a particular value.
- **Cumulative Distribution Function (CDF)**: denoted by $F_X(x)$, it gives the probability that the random variable takes on a value less than or equal to a certain value.

Probability

The most common random variable is the **standard normal random variable**, denoted by Z, which has a mean of 0 and a standard deviation of 1. Its PDF is given by:

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

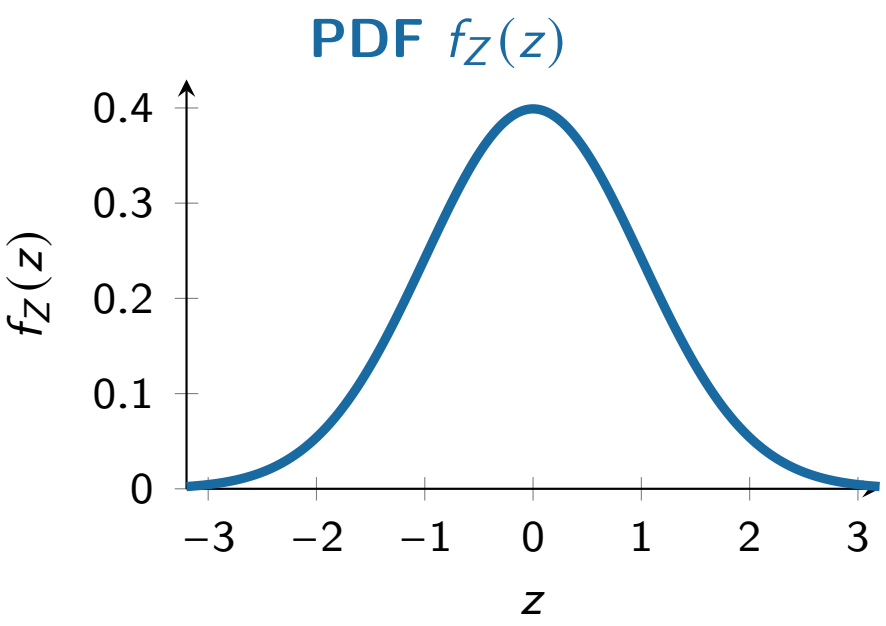
And its CDF is given by:

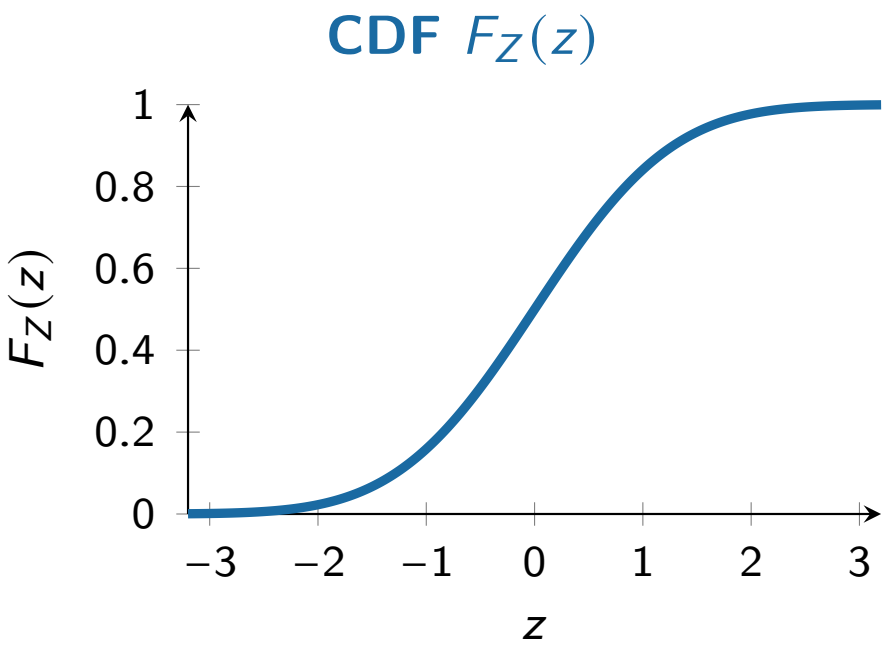
$$F_Z(z) = \frac{1}{2}\left(1 + \operatorname{erf}\left(\frac{z}{\sqrt{2}}\right)\right)$$

Where erf is the error function, which is a special function that arises in probability and statistics.

Probability

For a standard normal random variable, $Z \sim N(0,1)$, the PDF and CDF can be visualized as follows:





Probability

Another key concept in probability is the concept of **expected value**, which is crucial for derivatives pricing, as we will see later.

Expected Value

The expected value of a random variable is a measure of its central tendency, and it represents the average outcome we would expect if we were to repeat the random experiment many times. The expected value of a random variable X is defined as:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f_X(x) dx$$

For example, the expectation of a standard normal random variable Z is:

$$\mathbb{E}[Z] = 0$$

Probability

While the expected value measures the average outcome, the **variance** measures the variability or dispersion of the random variable around its expected value.

Variance

The variance of a random variable X is defined as:

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \int_{-\infty}^{\infty} (x - \mathbb{E}[X])^2 \cdot f_X(x)\,dx$$

For example, the variance of a standard normal random variable Z is:

$$\mathrm{Var}(Z) = 1$$

There is a key relationship between the expected value and the variance of a random variable, which is given by the following formula:

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Some useful properties of the expected value and variance that we will use later in derivatives pricing are the following:

- For a constant random variable $X = c$, we have $\mathbb{E}[X] = c$ and $\mathrm{Var}(X) = 0$.
- For a linear transformation of a random variable, $Y = aX + b$, we have $\mathbb{E}[Y] = a\mathbb{E}[X] + b$ and $\mathrm{Var}(Y) = a^2\mathrm{Var}(X)$;
- For the sum of two independent random variables, $Z = X + Y$, we have $\mathbb{E}[Z] = \mathbb{E}[X] + \mathbb{E}[Y]$ and $\mathrm{Var}(Z) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.

Probability

Mean and Variance Quick Calculations

For $Z \sim N(0,1)$ and $X \sim N(\mu, \sigma^2)$

Variable	**Mean**	**Variance**
Z	$\mathbb{E}[Z] = 0$	$\mathrm{Var}(Z) = 1$
X	$\mathbb{E}[X] = \mu$	$\mathrm{Var}(X) = \sigma^2$
Z^2	$\mathbb{E}[Z^2] = 1$	$\mathrm{Var}(Z^2) = \mathbb{E}[Z^4] = 2$

Useful identity: if $X = \mu + \sigma Z$ with $Z \sim N(0,1)$, then $X \sim N(\mu, \sigma^2)$.

Probability

Random variables allow us to model the future values of asset prices, interest rates, exchange rates, and other financial variables.

We use the subscript T to denote the time at which we are modeling the future value of these variables, for example:

- S_T: the future price of a stock at time T.
- L_T: the future spot interest rate at time T.
- R_T: the future return of an asset at time T.
- V_T: the future value of a derivative at time T.

And the subscript 0 to denote the current value of these variables, which is not random (i.e. S_0 is the current stock price, which is known and fixed).

Probability

The most common model for the future stock price, S_T, is the **Geometric Brownian Motion** (GBM), which assumes that the price follows a lognormal distribution[1]. The random variable S_T is expressed as:

$$S_T = S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z}$$

where:

- S_0 is the current stock price.
- r is the risk-free interest rate.
- σ is the volatility of the stock.
- Z is a standard normal random variable.
- T is the time to maturity.

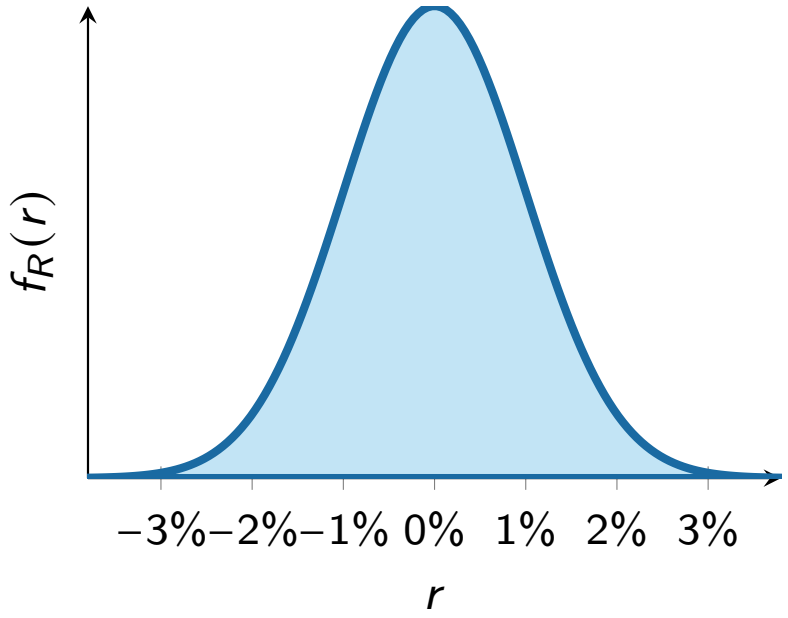
[1] The logarithm of a lognormal R.V is a normal R.V.

Probability

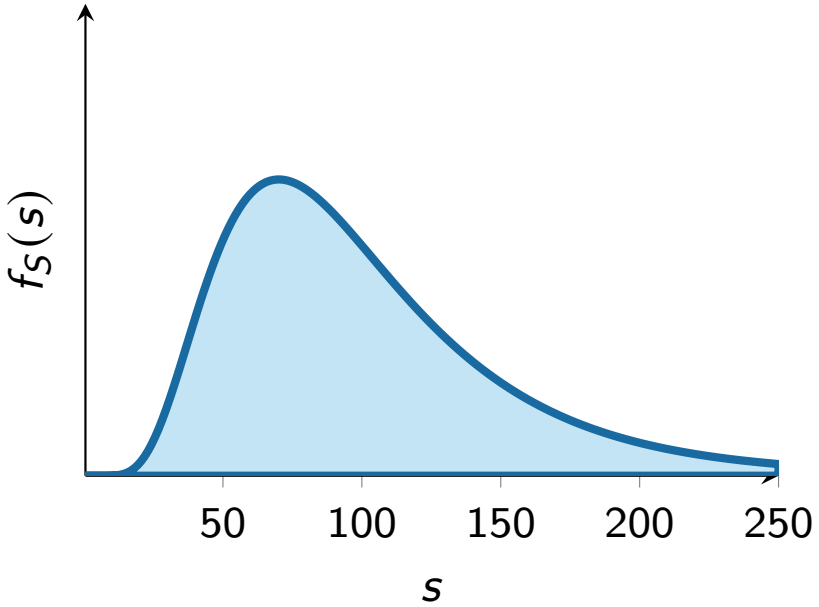
We can also model the future stock returns, $R_T = \ln(S_T/S_0)$, which follow a normal distribution:

$$R_T = (r - \frac{1}{2}\sigma^2)\,T + \sigma\sqrt{T}\,Z$$

Normal distribution



Log-Normal distribution



Probability

When a payoff is a function of a random variable, we use the following identity to compute its expectation.

Expectation of a Function

For a continuous random variable X with density $f_X(x)$ and any integrable function f:

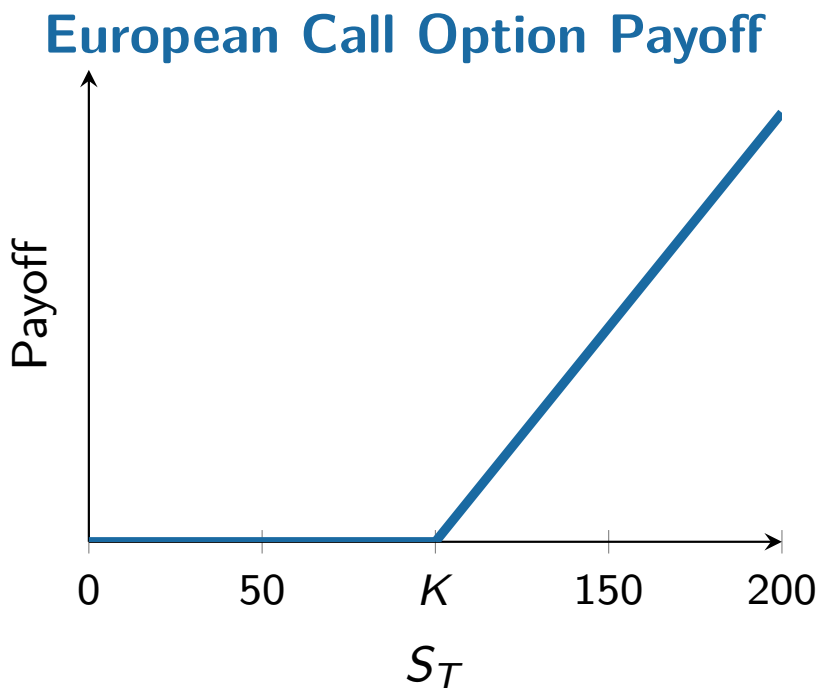
$$\mathbb{E}[f(X)] = \int_{-\infty}^{\infty} f(x)\, f_X(x)\, dx$$

This formula is extremely important in derivatives pricing, as derivatives are **functions of random variables**, and we need to compute their expected payoffs to determine their prices.

Probability

Let's say we have a European call option with strike price K and maturity T. The payoff function of this option is given by:

$$f(S_T) = \max(S_T - K, 0)$$



Probability

The price of this option can be calculated as the expected value of its payoff, discounted to the present by multiplying by the discount factor e^{-rT}, which accounts for the time value of money:

$$V_0 = e^{-rT}\mathbb{E}[\max(S_T - K, 0)]$$

This is the fundamental principle of derivatives pricing: **the price of a derivative is the present value of its expected payoff.**[2]

As long as we can model the probability distribution of the underlying asset price at maturity, we can calculate the expected payoff and thus the price of the derivative.

[2] Under the risk-neutral measure.

Probability

Pricing Formula

For any payoff function $f(x)$ and underlying asset X_T, the price of the derivative at time 0 can be calculated as:

$$V_0 = e^{-rT}\mathbb{E}[f(X_T)]$$

We can encounter two types of payoff functions:

- **Linear Payoffs**: These are payoffs that can be expressed as a linear function of the underlying asset price, such as $f(x) = ax + b$.
- **Convex Payoffs**: Non-linear payoffs that are convex functions of the underlying asset price, such as $f(x) = \max(x - K, 0)$ for a call option.

Probability

Some examples of linear payoffs include:

- A forward contract, which has a payoff of:

$$f(x) = x - F_0 \quad \text{where} \quad F_0 = S_0 e^{rT}$$

- A pure discount bond, which is bought at a discount and pays one at maturity:

$$f(x) = 1$$

- A coupon bearing bond, which pays periodic coupons and a final principal at maturity:

$$f(x) = \sum_{i=1}^{N} c + 1$$

Probability

- A floating rate note, which pays periodic coupons of variable rate x and a final principal at maturity:

$$f(x_i) = \sum_{i=1}^{N} x_i + 1$$

- An interest rate swap, which pays the difference between a fixed rate and a floating rate:

$$f(x_i) = \sum_{i=1}^{N} (x_i - c)$$

- An FX forward, which pays the difference between the spot exchange rate at maturity and a predetermined rate:

$$f(x) = x - K$$

Probability

Example 1: Pricing a pure discount bond

By using the pricing formula, we can calculate the price of a pure discount bond that pays 1 at maturity as follows:

$$V_0 = e^{-rT}\mathbb{E}[1] = e^{-rT}$$

Example 2: Pricing a Equity Forward Contract

By using the pricing formula, we can calculate the price of a forward contract on an equity, S_T, with forward price F_0 as follows:

$$V_0 = e^{-rT}\mathbb{E}[S_T - F_0] = e^{-rT}(\mathbb{E}[S_T] - F_0) = S_0 - F_0 e^{-rT} = 0$$

Probability

Example 3: Pricing a coupon bearing bond

Here, we discount each cash flow separately, since they occur at different times. We assume a constant discount rate r for simplicity:

$$V_0 = \mathbb{E}\left[\sum_{i=1}^{N} ce^{-rt_i} + e^{-rT}\right] = \sum_{i=1}^{N} ce^{-rt_i} + e^{-rT}$$

Example 4: Pricing a floating rate note

The floating rate note pays at each time t_i a coupon equal to the floating rate L_i, which is a random variable:

$$V_0 = \mathbb{E}\left[\sum_{i=1}^{N} L_i e^{-rt_i} + e^{-rT}\right] = \sum_{i=1}^{N} \mathbb{E}[L_i] e^{-rt_i} + e^{-rT}$$

Probability

One key property of linear payoffs is the following:

Linearity of Expectation

For any linear payoff function $f(x) = ax + b$ and random variable X, the expected value of the payoff can be calculated as:

$$\mathbb{E}[f(X)] = a\mathbb{E}[X] + b$$

In other words, the expectation of the function of a random variable is equal to the function of the expectation of the random variable, when the function is linear.

$$\mathbb{E}[f(X)] = f(\mathbb{E}[X])$$

Probability

By making use of this property, **we avoid the need to calculate the entire probability distribution** of X to find the expected payoff, and we can simply calculate the expected value of X and then apply the linear function to it.

For example, when we priced the forward contract, we computed the payoff as:

$$\mathbb{E}[S_T - F_0] = \mathbb{E}[S_T] - F_0 = S_0 e^{rT} - F_0$$

Avoiding the need to compute the entire distribution of S_T.

Probability

Under the adequate probability measure[3], the expectation of a random variable such as a stock price or rate is equal to its **forward price**:

- A future stock price S_T is a forward price, when taking its expectation, $\mathbb{E}^{\mathbb{Q}}[S_T]$, under the risk-neutral measure $\mathbb{Q}$.
- A future spot interest rate L_T is a forward rate, when taking its expectation, $\mathbb{E}^{\mathbb{T}}[L_T]$ under the T-forward measure.
- A future exchange rate X_T is a forward exchange rate, when taking its expectation, $\mathbb{E}^{\mathbb{Q}}[X_T]$, under the risk-neutral measure $\mathbb{Q}$.

Usually, the forward price (or rate) is observed in the market or is easy to calculate, so we can use it directly to price derivatives with linear payoffs.

[3] A probability measure is a mathematical function that assigns probabilities to events. In derivatives pricing we use probability measures that rule out arbitrage opportunities.

Probability

Moving to convex payoffs, here are some examples:

- A European call option, which has a payoff of:

$$f(x) = \max(x - K, 0)$$

- A European put option, which has a payoff of:

$$f(x) = \max(K - x, 0)$$

- A digital option, which pays a fixed amount if the underlying asset price is above or below a certain level at maturity:

$$f(x) = \begin{cases} 1 & \text{if } x > K \\ 0 & \text{otherwise} \end{cases}$$

Probability

- An accrual note which pays a variable amount based on an accrual condition, g, which is a convex function of an underlying rate x:

$$f(x_i) = \sum_{i=1}^{N} g(x_i) + 1$$

- A CMS spread option, which pays the difference between two swap rates if it is positive at maturity:

$$f(x, y) = \max(x - y, 0)$$

- A hybrid structured note that pays a convex function of an interest rate, $h(x)$, conditional to the performance of an stock, $g(y)$:

$$f(x_i, y_i) = \sum_{i=1}^{N} h(x_i) \cdot g(y_i) + 1$$

Probability

- A worst-of basket option, which pays the worst performance of a basket of assets at maturity:

$$f(x_1, x_2, ..., x_n) = \min(x_1, x_2, ..., x_n)$$

- An asian option, which pays the average price of the underlying asset over a certain period:

$$f(x) = \max\left(\frac{1}{n}\sum_{i=1}^{n} x_i - K, 0\right)$$

- A caplet or floorlet, which pays the difference between a floating rate and a fixed rate if it is above or below a certain level at maturity:

$$f(x) = \begin{cases} \Delta t \cdot \max(x - K, 0) & \text{for a caplet} \\ \Delta t \cdot \max(K - x, 0) & \text{for a floorlet} \end{cases}$$

Probability

For convex payoffs, **we cannot use the linearity of expectation**, and we need to calculate the entire probability distribution of the underlying asset price at maturity to find the expected payoff.

Example 5: Pricing a call option

The price of a European call option can be calculated as the expected value of its payoff, discounted to the present:

$$C_0 = e^{-rT}\mathbb{E}[\max(S_T - K, 0)] = e^{-rT}\int_K^\infty (s-K) f_{S_T}(s)\,ds$$

where $f_{S_T}(s)$ is the probability density function of the stock price at maturity.

Probability

If we try to apply the linearity of expectation to this payoff, we would get:

$$\mathbb{E}[\max(S_T - K, 0)] \neq \max(\mathbb{E}[S_T] - K, 0)$$

Which is not correct and would lead to a zero price for the call option if

$$\mathbb{E}[S_T] = S_0 e^{rT} \leq K$$

Implying that the option is worthless if not ITM, for any expiry date, so we can buy ATM or options for free and wait until they become ITM, which is a clear arbitrage opportunity.

Intuitively, we would suspect that the following inequality should hold:

$$\mathbb{E}[\max(S_T - K, 0)] \geq \max(\mathbb{E}[S_T] - K, 0)$$

Probability

This is indeed the case, and it is a consequence of the convexity of the payoff function. This property is known as **Jensen's Inequality**:

Jensen's Inequality

For any convex function f and random variable X, the following inequality holds:

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$$

In other words, the expectation of a convex function of a random variable is greater than or equal to the function of the expectation of the random variable.

Probability

Another key concept is **Convexity Adjustments**. We can price a derivative with a convex payoff by using the following formula:

$$V_0 = e^{-rT}\mathbb{E}[f(X_T)] = e^{-rT}f(\mathbb{E}[X_T]) + \mathcal{C}$$

Where $\mathcal{C}$ is the convexity adjustment. Here we have computed the price of the derivative as if it had a linear payoff, $f(\mathbb{E}[X_T])$, and then we have added a convexity adjustment to account for the fact that the payoff is actually convex.

The convexity adjustment should be positive and needs to be computed by simulating the underlying asset price at maturity, X_T.

Probability

Another type of convexity adjustment is the one that arises when we want to take the expectation of a product of two dependent random variables, X and Y:

$$\mathbb{E}[X \cdot Y] \neq \mathbb{E}[X] \cdot \mathbb{E}[Y]$$

This is due to the fact that there exists a correlation between X and Y, and the expected value should be computed as:

$$\mathbb{E}[X \cdot Y] = \mathbb{E}[X] \cdot \mathbb{E}[Y] + \rho \cdot \sigma_X \cdot \sigma_Y$$

Where ρ is the correlation between X and Y, and σ_X and σ_Y are the standard deviations of X and Y, respectively.

Probability

In the previous example, we can identify another type of convexity adjustment:

$$C = \rho \cdot \sigma_X \cdot \sigma_Y$$

So the expectation of the product of two random variables can be expressed as the product of their expectations plus a convexity adjustment that accounts for the correlation between them.

$$\mathbb{E}[X \cdot Y] = \mathbb{E}[X] \cdot \mathbb{E}[Y] + C$$

Here C can be positive or negative, depending on the sign of the correlation ρ between X and Y.

Calculus

Calculus is another fundamental branch of mathematics for derivatives pricing, as it allows us to compute sensitivities, solve differential equations, and perform optimization.

In derivatives pricing, we often use calculus to:

- Compute the Greeks, which are the sensitivities of the derivative price to changes in underlying parameters.
- Solve Ordinary Differential Equations (ODEs) and derive Partial Differential Equations (PDEs) that arise in the pricing of derivatives.
- Impose no arbitrage conditions.

Calculus

The most basic concept in calculus is the **derivative**, which measures the rate of change of a function with respect to its variables.

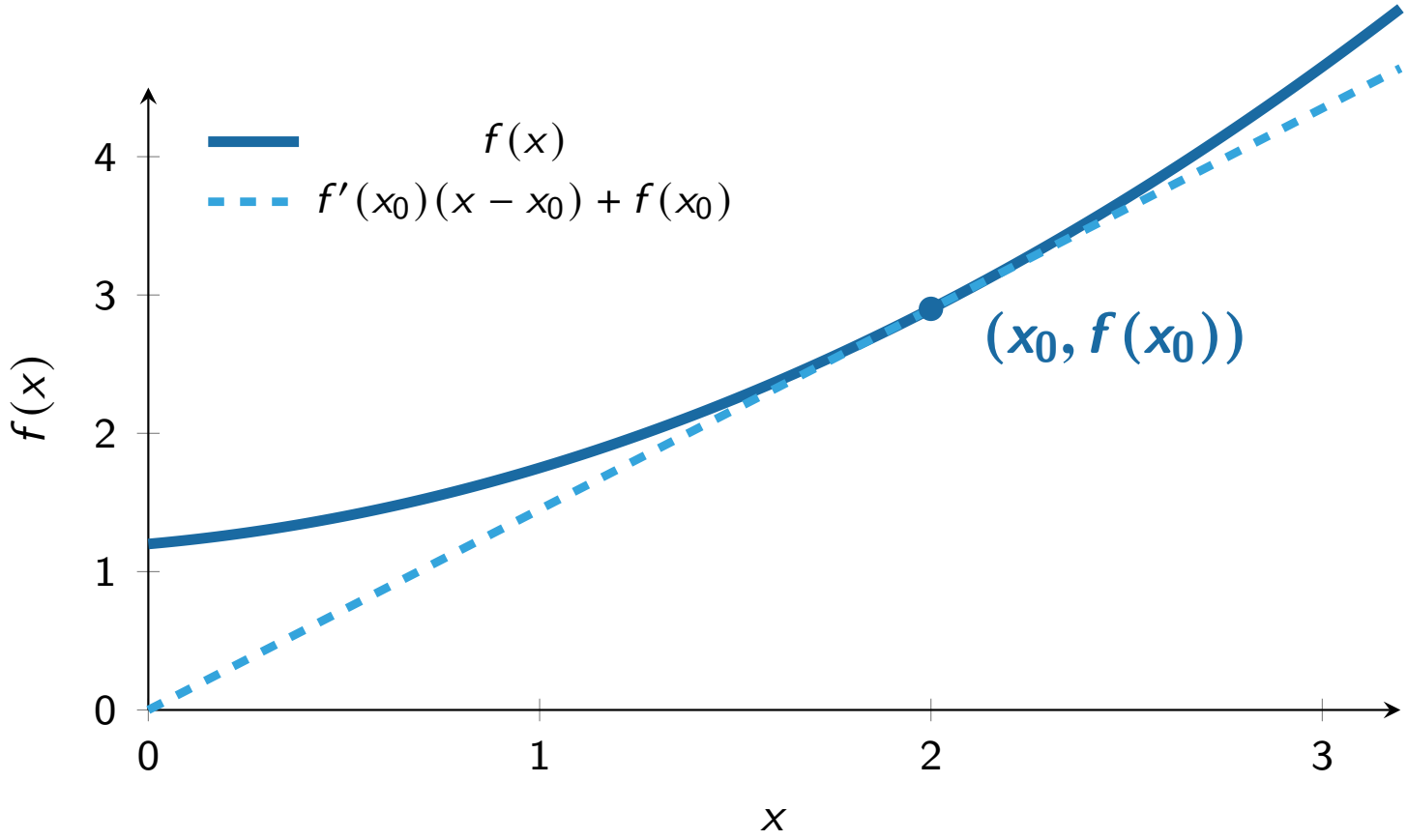
Derivative

The derivative of a function $f(x)$ with respect to x is defined as:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

The derivative gives us information about the slope of the function at any point, and it is a fundamental tool for understanding how the price of a derivative changes with respect to changes in the underlying risk factors.

Calculus



The derivative at x_0 is the slope of the line that best approximates $f(x)$ locally.

Calculus

When the function has two or more variables, we can use the concept of **partial derivatives** to compute the rate of change of the function with respect to each variable while keeping the others constant.

Partial Derivative

The partial derivative of a function $f(x, y)$ with respect to x is defined as:

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$$

And the partial derivative with respect to y is defined as:

$$\frac{\partial f}{\partial y} = \lim_{h \to 0} \frac{f(x, y+h) - f(x, y)}{h}$$

Calculus

Another fundamental concept in calculus is the **Taylor expansion**, which allows us to approximate a function around a certain point by using its derivatives at that point.

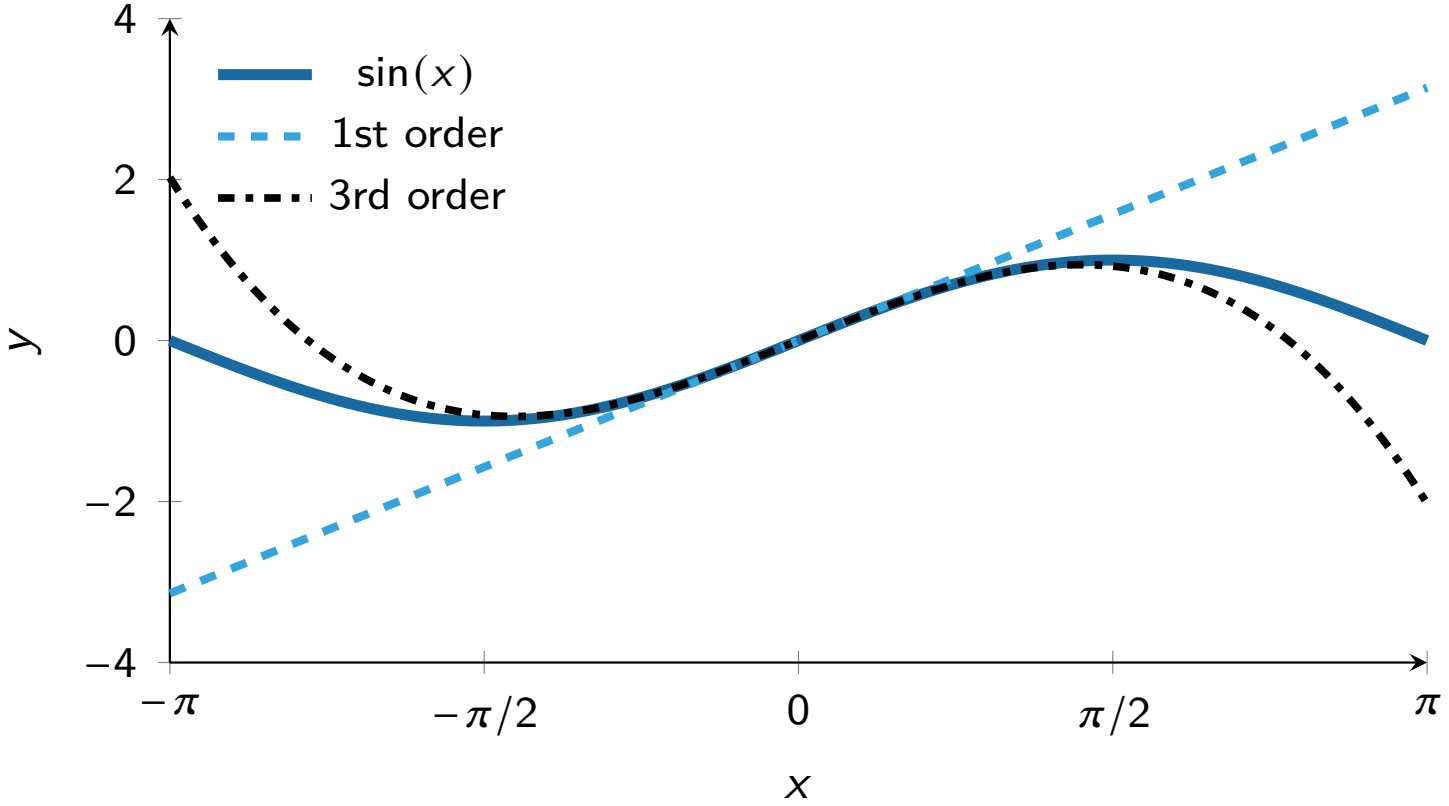
Taylor Expansion

The Taylor expansion of a function $f(x)$ around a point x_0 is given by:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots$$

The more terms we include in the expansion, the better the approximation will be, especially for values of x close to x_0.

Taylor Approximation of a Function



Calculus

The Taylor expansion can also be applied to functions of multiple variables, and it allows us to approximate the function around a certain point by using its partial derivatives at that point.

Multivariate Taylor Expansion

The Taylor expansion of a function $f(x, y)$ around a point (x_0, y_0) is given by:

$$f(x, y) = f(x_0, y_0) + \frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0) + \cdots$$

Calculus

The Taylor expansion is widely used in derivatives pricing to compute sensitivities and to derive pricing formulas.

For example, we can use the Taylor expansion to derive the change in the price of a derivative with respect to changes in the underlying risk factors:

$$\Delta V \approx \frac{\partial V}{\partial S}\Delta S + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}(\Delta S)^2 + \frac{\partial V}{\partial \sigma}\Delta\sigma + \frac{\partial V}{\partial t}\Delta t + \cdots$$

Where:

- $\Delta V = V(S + \Delta S, \sigma + \Delta\sigma, t + \Delta t) - V(S, \sigma, t)$ is the change in the derivative price.
- ΔS, $\Delta\sigma$, and Δt are the changes in the underlying price, volatility, and time to maturity, respectively.

Calculus

In the previous example, the partial derivatives of the derivative price V with respect to the underlying risk factors are known as the **Greeks**, which are key sensitivities in derivatives pricing:

Main Greeks

$\Delta = \frac{\partial V}{\partial S}$	Sensitivity of option price to the underlying price.
$\Gamma = \frac{\partial^2 V}{\partial S^2}$	Sensitivity of Delta to the underlying price.
$\nu = \frac{\partial V}{\partial \sigma}$	Sensitivity of option price to volatility.
$\Theta = \frac{\partial V}{\partial t}$	Sensitivity of option price to the passage of time.

Calculus

There is another use of partial derivatives in finance which is to impose no-arbitrage conditions in the price of derivatives. For example, in the case of a European call option [4], we have the following no-arbitrage conditions:

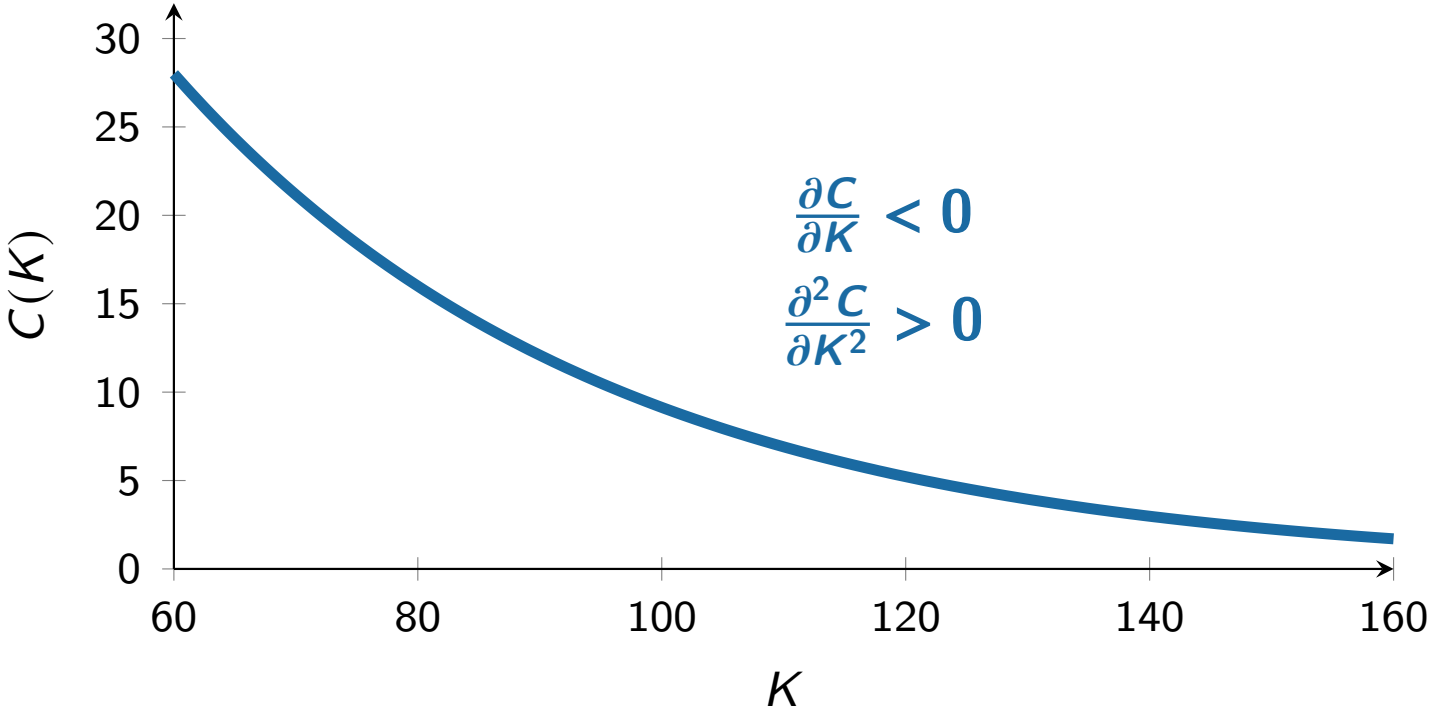
No-Arbitrage Conditions for a Call Option

$\frac{\partial C}{\partial K} < 0$	Call price decreases as strike K increases.
$\frac{\partial^2 C}{\partial K^2} > 0$	Call price is convex with respect to strike K.
$\frac{\partial C}{\partial T} > 0$	Call price increases as maturity T increases.

[4] We usually denote C for call option and V for a generic derivative.

Calculus

Call Price as a Function of Strike



Calculus

Recall that the price of a European call option can be calculated as:

$$C(K) = e^{-rT}\mathbb{E}[\max(S_T - K, 0)] = e^{-rT}\int_{-\infty}^{\infty} \max(S_T - K, 0) \cdot f_{S_T}(s)ds$$

If we apply the no arbitrage conditions to this formula, we have that:

Breeden-Litzenberger Theorem

If $C(K)$ is the price of a European call with strike K and maturity T, then:

$$\frac{\partial C(K)}{\partial K} = e^{-rT}\left(F_{S_T}(K) - 1\right) < 0$$

$$\frac{\partial^2 C(K)}{\partial K^2} = e^{-rT} f_{S_T}(K) > 0$$

Calculus

The Breeden-Litzenberger Theorem allows us to connect the shape of the call price curve as a function of strike to the risk-neutral distribution of the underlying asset price at maturity.

Price Restrictions	**Probabilistic Restrictions**
$\frac{\partial C}{\partial K} = e^{-rT}(F_{S_T}(K) - 1) < 0$	The CDF must be less than 1.
$\frac{\partial^2 C}{\partial K^2} = e^{-rT} f_{S_T}(K) > 0$	The PDF must be positive.

Shape restrictions on call prices are equivalent to probabilistic restrictions on the risk-neutral distribution of S_T.

Calculus

The next key concept in calculus is the **integral**, which is the inverse operation of differentiation and allows us to compute areas under curves, which is essential for calculating expected values and probabilities.

Integral

The integral of a function $f(x)$ from a to b is defined as:

$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

Where $\Delta x = \frac{b-a}{n}$ and $x_i^* = a + i\Delta x$ is a sample point in the i-th subinterval.

Calculus

Integrals and derivatives are closely related, and they are connected by the Fundamental Theorem of Calculus, which states that differentiation and integration are inverse operations.

Fundamental Theorem of Calculus

If f is a continuous function on the interval $[a, b]$ and F is an antiderivative of f on that interval, then:

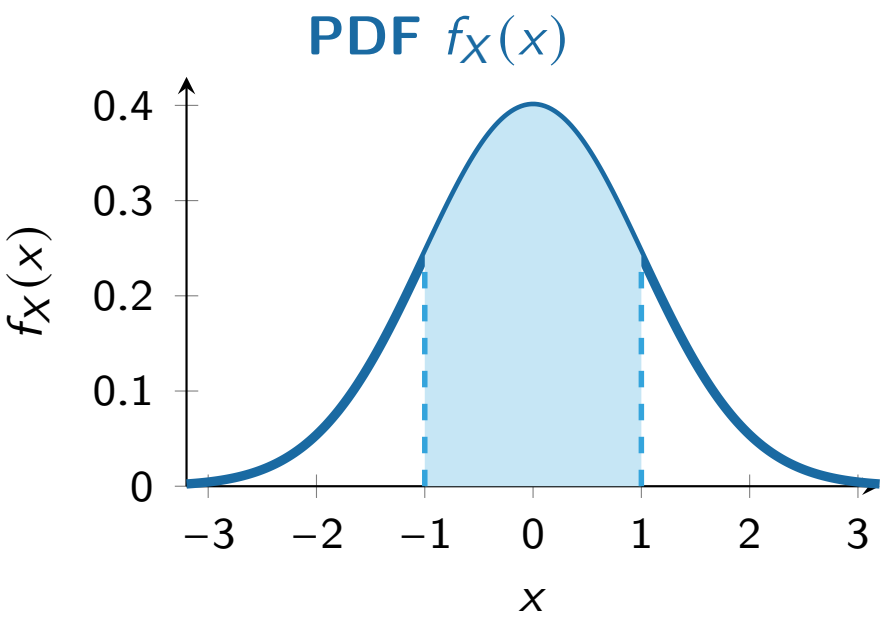
$$\int_a^b f(x)dx = F(b) - F(a)$$

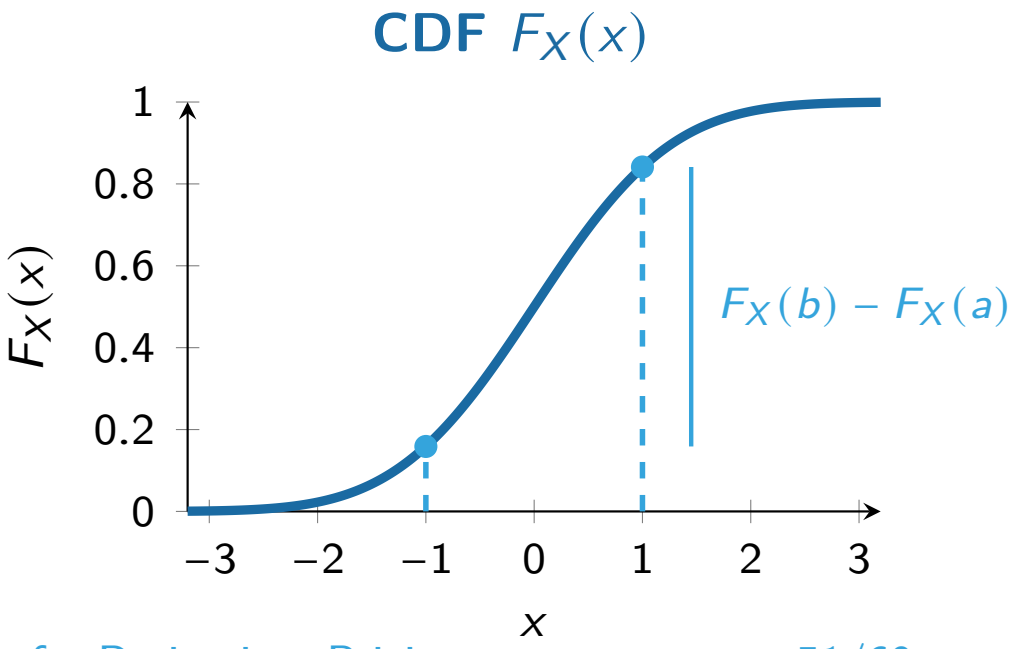
Calculus

Relationship Between PDF and CDF

The PDF and the CDF of a random variable are connected by the fundamental theorem of calculus:

$$\int_a^b f_X(x)\,dx = F_X(b) - F_X(a), \qquad a < b.$$





Calculus

Ordinary Differential Equations (ODEs) are equations that involve derivatives of a function with respect to a single variable. One example of ODE is a bank account that grows at a constant interest rate r:

$$dB(t) = rB(t)dt$$

We can solve this ODE by separating variables and integrating both sides:

$$\int \frac{1}{B(t)} dB(t) = \int r dt$$

This gives us:

$$B(t) = e^{rt+C} = B(0)e^{rt}$$

Where $B(0) = e^C$ is the initial value of the bank account at time $t = 0$.

Calculus

Partial Differential Equations (PDEs) are equations that involve derivatives of a function with respect to multiple variables. In finance, we use them to price derivatives that depend on multiple risk factors, such as the underlying asset price and time.

One example of PDE is the Black-Scholes equation for a European call option:

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS\frac{\partial C}{\partial S} - rC = 0$$

Which gives the same price as the expected value of the payoff under the risk-neutral measure, but it is derived using a different approach.

Stochastic Calculus

Stochastic Differential Equations (SDEs) are equations that involve stochastic processes, which are random variables that evolve over time. They are used to model the dynamics of underlying asset prices and interest rates in derivatives pricing.

One example of SDE is the Geometric Brownian Motion (GBM) model for an asset price S_t:

$$dS_t = rS_t dt + \sigma S_t dW_t$$

Where r is the risk-free rate, σ is the volatility, and W_t is a standard Brownian motion.

Stochastic Calculus

Note the similarities between the ODE for the bank account and the SDE for the asset price.

$$\underbrace{\frac{dS_t}{S_t}}_{\text{Change in stock returns}} = \underbrace{rdt}_{\text{Deterministic expected return}} + \underbrace{\sigma dW_t}_{\text{Random shock}}$$

The main difference is that the SDE includes a stochastic term, $\sigma S_t dW_t$, which introduces randomness into the evolution of the asset price.

Stochastic Calculus

In finance, there are three ways to model the dynamics of a variable:

Constant

r

Same value at every point in time.

Deterministic

$r(t)$

Evolves through a known time function.

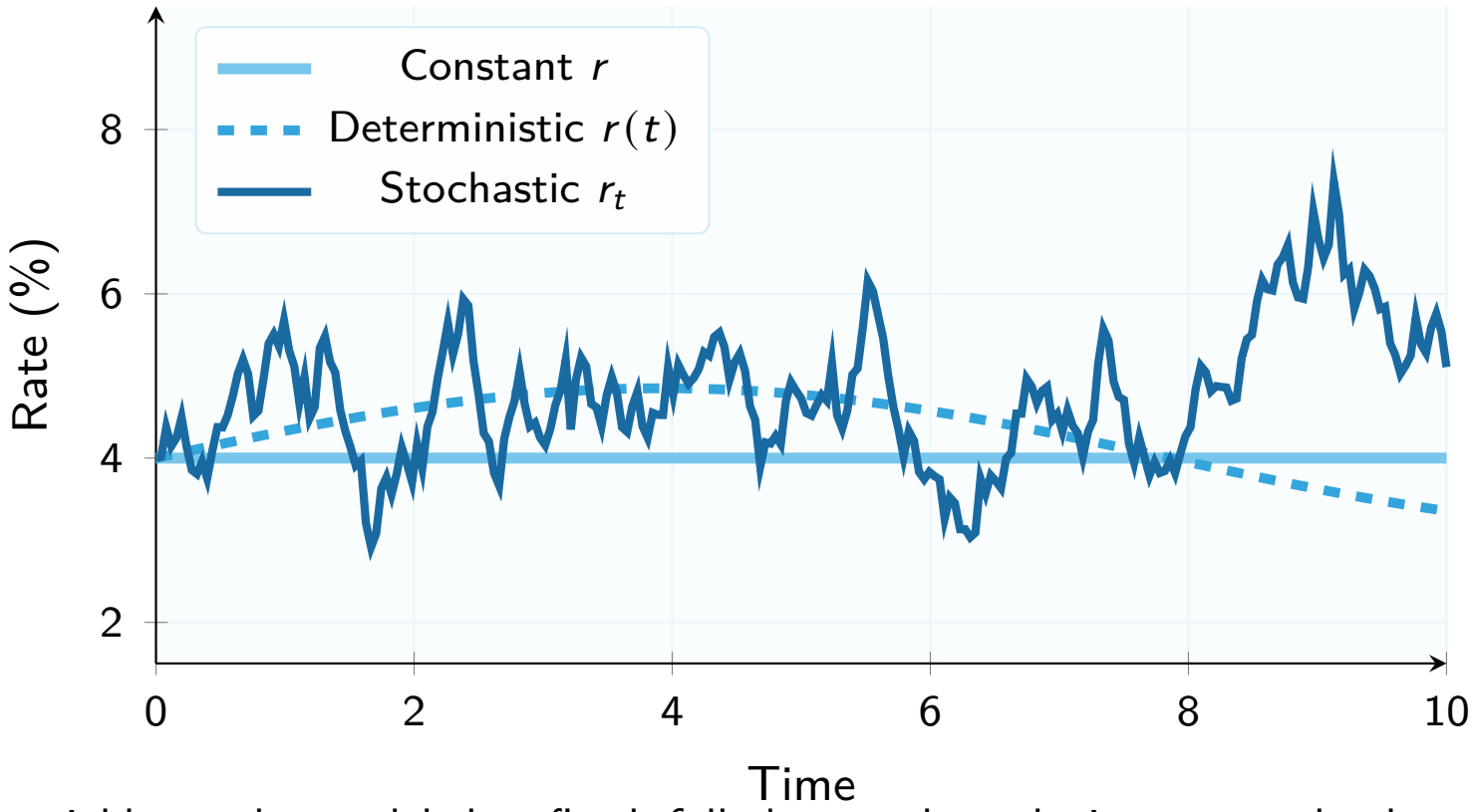
Stochastic

r_t

Evolves randomly through time.

Example: the risk-free rate can be modeled as a constant, a deterministic curve, or a stochastic process.

Examples of Interest-Rate Dynamics



The same variable can be modeled as fixed, fully known through time, or randomly evolving.

Stochastic Calculus

We have previously seen that PDEs and discounted expectations are two different approaches to price derivatives, but they are closely related. The solution of:

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS\frac{\partial C}{\partial S} - rC = 0$$

Is actually the same as the expected value of the payoff under the risk-neutral measure:

$$C = e^{-rT}\mathbb{E}[\max(S_T - K, 0)] = e^{-rT}\int_{-\infty}^{\infty} \max(s - K, 0) f_{S_T}(s)\, ds$$

Stochastic Calculus

The connection between PDEs and discounted expectations is given by the **Feynman-Kac formula**, which states that the solution of a certain class of PDEs can be expressed as the expected value of a stochastic process.

Feynman-Kac Formula

If $C(t, S)$ is the solution of the PDE:

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS\frac{\partial C}{\partial S} - rC = 0$$

with terminal condition $C(T, S) = \max(S - K, 0)$, then $C(t, S)$ can be expressed as:

$$C(t, S) = e^{-r(T-t)}\mathbb{E}[\max(S_T - K, 0)]$$

Conditioned on $S_t = S$ and where S_t follows the GBM under the RNM.

Numerical Methods

In practice, we often need to use textcolormydarkblue***Numerical Methods*** to price derivatives, especially when the underlying models are complex and do not have closed-form solutions. Some common numerical methods include:

- **Monte Carlo Simulation**: A method that uses random sampling to estimate the expected value of a derivative's payoff under the risk-neutral measure.
- **Finite Difference Methods**: A method that enables the computation of Greeks and the price of derivatives by discretizing the PDEs that arise in derivatives pricing.

Numerical Methods

Monte Carlo Simulation is a powerful and flexible method for pricing derivatives, especially when the underlying model is complex and does not have a closed-form solution.

Recall that the price of a derivative can be expressed as the expected value of its payoff under the risk-neutral measure:

$$V_0 = e^{-rT}\mathbb{E}[f(X_T)]$$

Where X_T is the underlying asset price at maturity and $f(X_T)$ is the payoff of the derivative. When there is no close form solution for this expectation, we can use Monte Carlo simulation to estimate it.

Numerical Methods

Monte Carlo Estimator

We can approximate the price of the derivative, V_0, by simulating N paths of the underlying asset price at maturity, X_T^i, and averaging the discounted payoff across those paths:

$$V_0 \approx e^{-rT} \frac{1}{N} \sum_{i=1}^{N} f(X_T^i)$$

Here, X_T^i is the simulated asset price at maturity for the i-th path. The accuracy of this approximation depends on the number of paths N and the variance of the payoff function $f(X_T)$.

Numerical Methods

The **Monte Carlo Error** is the difference between the true price of the derivative and the estimated price obtained from the Monte Carlo simulation. It can be expressed as:

$$\text{Error} = |V_0 - \hat{V}_0|$$

Where V_0 is the true price of the derivative and $\hat{V}_0$ is the estimated price from the Monte Carlo simulation. The error can be reduced by increasing the number of simulated paths N:

$$\text{Error} \propto \frac{1}{\sqrt{N}}$$

For example, if we want to reduce the error by a factor of 2, we need to increase the number of paths by a factor of 4.

Numerical Methods

Whenever we need to compute the Greeks and there is no closed-form solution, we can use the **finite difference method** to approximate the derivatives of the price with respect to the underlying risk factors.

Finite Difference Method (FDM)

The FDM approximates the derivative of a function by using the values of the function at nearby points. The first derivative can be approximated as:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

Where h is a small perturbation. Similarly, the second derivative can be approximated as:

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

Numerical Methods

The finite difference approximation for the Delta of a derivative:

$$\Delta \approx \frac{V(S+h) - V(S-h)}{2h}$$

Where h is a small perturbation in the underlying price S. Similarly, the Gamma can be approximated as:

$$\Gamma \approx \frac{V(S+h) - 2V(S) + V(S-h)}{h^2}$$

And the Vega can be approximated as:

$$\nu \approx \frac{V(\sigma+h) - V(\sigma-h)}{2h}$$

Where h is a small perturbation in the volatility σ.

Linear Algebra

Lastly, ***Linear Algebra*** is also an important branch of mathematics for derivatives pricing, as it allows us to work with vectors and matrices, which are essential for modeling multiple risk factors and for performing numerical computations.

One key concept is the **Cholesky decomposition**, which is a method for decomposing a positive-definite matrix into the product of a lower triangular matrix and its transpose.

This is particularly useful for simulating correlated random variables, which is common in derivatives pricing when we have multiple underlying assets or risk factors.

Linear Algebra

Cholesky Decomposition

If A is a positive-definite matrix, then there exists a unique lower triangular matrix L such that:

$$A = LL^T$$

Where L^T is the transpose of L.

This decomposition allows us to generate correlated random variables by applying the transformation LZ, where Z is a vector of independent standard normal random variables.

Linear Algebra

For example, we can use the Cholesky decomposition to generate correlated random variables for two assets with a correlation ρ:

1. Suppose we have two assets with a correlation ρ. We can construct the covariance matrix:
$$\Sigma = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$$
2. Perform the Cholesky decomposition to find the lower triangular matrix L such that $\Sigma = LL^T$.
3. Generate correlated random variables by multiplying L with a vector of independent standard normal random variables, $Z = (Z_1, Z_2)^T$:
$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = L \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix}$$

Next Steps

If you enjoyed this introduction to the mathematical tools used in derivatives pricing, click the links below to explore more in-depth materials on each topic:

- ***Stochastic Calculus***: A comprehensive introduction to stochastic calculus and its applications in finance.
- ***Risk-Neutral Pricing***: An in-depth look at risk-neutral pricing in discrete and continuous time.